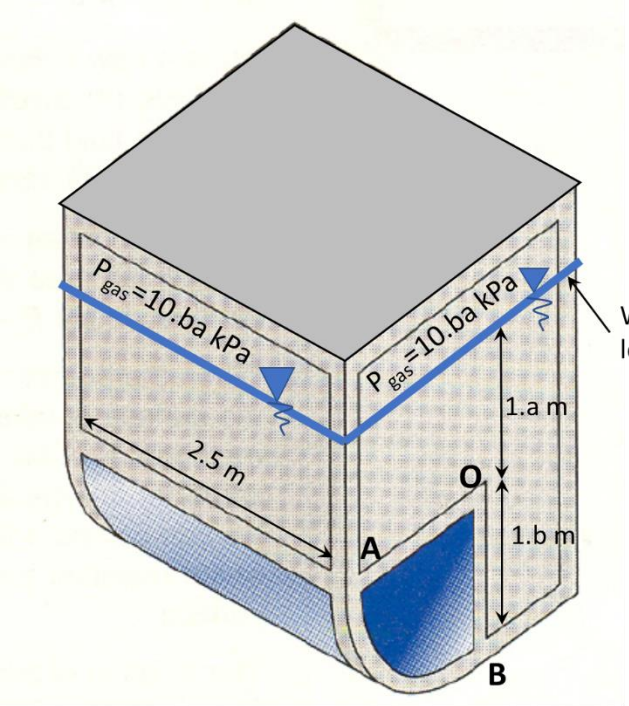
2022-2023 SPRING SEMESTER FLUID MECHANICS MIDTERM1, Date:25/04/2023

Name: **Surname:** **Student ID:** _ _ _ _ _ _ _ _ _ _
$\underbrace{}_{a}\ \underbrace{}_{b}$

QUESTION 4 (25p): The tank shown in the figure is partly filled with water and pressurized by CO_2.

a) Calculate the pressure force acting on the quarter-circular plane surface AOB.
b) Determine the application point of the force.
c) Draw hydrostatic pressure distribution over the plane surface AOB.



a and b are the last two digits of your student ID.
Example: Student ID: 17042082 a=8 b=2
(a and b are the digits after the comma)

$$h_{\substack{G,quarter\\circle}} = \frac{4r}{3\pi}$$

$$I_{\substack{G,quarter\\circle}} = \frac{\pi r^4}{16}$$